

PRACTICAL COURSE WORKBOOK

Biomedical Data Analysis with Python

Work with digital-health information responsibly

LEVEL	Intermediate
GUIDED TIME	5 hours across 12 lessons
PROJECT	a safe, fictional health-data workflow and review
SKILLS	BioPython, Genomics, Pandas, Clinical data, R, Critical evaluation
EDITION	Structured draft - version 0.9

WHAT YOU WILL BE ABLE TO DO

- Use BioPython appropriately in a realistic, bounded task.
- Use Genomics appropriately in a realistic, bounded task.
- Use Pandas appropriately in a realistic, bounded task.
- Use Clinical data appropriately in a realistic, bounded task.

WHO THIS IS FOR

Learners exploring health technology without replacing clinical expertise.

BEFORE YOU BEGIN

- Basic data literacy

Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

WORKBOOK GUIDE

A clear route through the course

Biomedical Data Analysis with Python is a structured, practical course covering BioPython, Genomics, Pandas, Clinical data, R. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 BioPython: Health context	1. Care pathways with BioPython 2. Health data with Genomics 3. Stakeholders with Pandas
02 Genomics: Digital systems	1. Standards with Clinical data 2. Interoperability with R 3. Workflows with BioPython
03 Pandas: Quality and safety	1. Validation with Genomics 2. Privacy with Pandas 3. Human oversight with Clinical data
04 Clinical data: Applied evaluation	1. Implementation with R 2. Monitoring with BioPython 3. Equity with Genomics

HOW TO USE EACH LESSON PAGE

- 1 Read the objective and identify the decision it supports.
- 2 Attempt the retrieval prompt before checking the online explanation.
- 3 Reproduce the worked example with the stated input.
- 4 Test one normal case and one boundary or failure case.
- 5 Record the result, limitation and next action in the evidence log.

PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

MODULE 01

BioPython: Health context

Apply BioPython through care pathways, health data, stakeholders.

LESSON 01 / 18 MIN

Care pathways with BioPython

Learning objectives	• Explain care pathways in the context of Biomedical Data Analysis with Python. • Apply BioPython to a bounded practical task. • Evaluate the result using explicit quality criteria.
Key terms	care pathways, BioPython, healthcare
Retrieval prompt	In Biomedical Data Analysis with Python, which evidence best supports a care pathways result produced with BioPython?
Evidence to keep	☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 02 / 22 MIN

Health data with Genomics

Learning objectives	• Explain health data in the context of Biomedical Data Analysis with Python. • Apply Genomics to a bounded practical task. • Evaluate the result using explicit quality criteria.
Key terms	health data, Genomics, healthcare
Retrieval prompt	In Biomedical Data Analysis with Python, which evidence best supports a health data result produced with Genomics?
Evidence to keep	☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 03 / 26 MIN

Stakeholders with Pandas

Learning objectives	• Explain stakeholders in the context of Biomedical Data Analysis with Python. • Apply Pandas to a bounded practical task. • Evaluate the result using explicit quality criteria.
Key terms	stakeholders, Pandas, healthcare
Retrieval prompt	In Biomedical Data Analysis with Python, which evidence best supports a stakeholders result produced with Pandas?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 02

Genomics: Digital systems

Apply Genomics through standards, interoperability, workflows.

LESSON 04 / 30 MIN

Standards with Clinical data

Learning objectives

- Explain standards in the context of Biomedical Data Analysis with Python.
- Apply Clinical data to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

standards, Clinical data, healthcare

Retrieval prompt

In Biomedical Data Analysis with Python, which evidence best supports a standards result produced with Clinical data?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 05 / 18 MIN

Interoperability with R

Learning objectives

- Explain interoperability in the context of Biomedical Data Analysis with Python.
- Apply R to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

interoperability, R, healthcare

Retrieval prompt

In Biomedical Data Analysis with Python, which evidence best supports a interoperability result produced with R?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 06 / 22 MIN

Workflows with BioPython

Learning objectives

- Explain workflows in the context of Biomedical Data Analysis with Python.
- Apply BioPython to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

workflows, BioPython, healthcare

Retrieval prompt

In Biomedical Data Analysis with Python, which evidence best supports a workflows result produced with BioPython?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 03

Pandas: Quality and safety

Apply Pandas through validation, privacy, human oversight.

LESSON 07 / 26 MIN

Validation with Genomics

Learning objectives	• Explain validation in the context of Biomedical Data Analysis with Python. • Apply Genomics to a bounded practical task. • Evaluate the result using explicit quality criteria.
Key terms	validation, Genomics, healthcare
Retrieval prompt	In Biomedical Data Analysis with Python, which evidence best supports a validation result produced with Genomics?
Evidence to keep	☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 08 / 30 MIN

Privacy with Pandas

Learning objectives	• Explain privacy in the context of Biomedical Data Analysis with Python. • Apply Pandas to a bounded practical task. • Evaluate the result using explicit quality criteria.
Key terms	privacy, Pandas, healthcare
Retrieval prompt	In Biomedical Data Analysis with Python, which evidence best supports a privacy result produced with Pandas?
Evidence to keep	☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 09 / 18 MIN

Human oversight with Clinical data

Learning objectives	• Explain human oversight in the context of Biomedical Data Analysis with Python. • Apply Clinical data to a bounded practical task. • Evaluate the result using explicit quality criteria.
Key terms	human oversight, Clinical data, healthcare
Retrieval prompt	In Biomedical Data Analysis with Python, which evidence best supports a human oversight result produced with Clinical data?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 04

Clinical data: Applied evaluation

Apply Clinical data through implementation, monitoring, equity.

LESSON 10 / 22 MIN

Implementation with R

Learning objectives

- Explain implementation in the context of Biomedical Data Analysis with Python.
- Apply R to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

implementation, R, healthcare

Retrieval prompt

In Biomedical Data Analysis with Python, which evidence best supports a implementation result produced with R?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 11 / 26 MIN

Monitoring with BioPython

Learning objectives

- Explain monitoring in the context of Biomedical Data Analysis with Python.
- Apply BioPython to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

monitoring, BioPython, healthcare

Retrieval prompt

In Biomedical Data Analysis with Python, which evidence best supports a monitoring result produced with BioPython?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 12 / 30 MIN

Equity with Genomics

Learning objectives

- Explain equity in the context of Biomedical Data Analysis with Python.
- Apply Genomics to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

equity, Genomics, healthcare

Retrieval prompt

In Biomedical Data Analysis with Python, which evidence best supports a equity result produced with Genomics?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

CAPSTONE PROJECT

Produce a reviewable Biomedical Data Analysis with Python project using BioPython, Genomics, Pandas.

DELIVERABLE	A working Biomedical Data Analysis with Python artefact plus an evidence-based self-review.
TOOLS	A suitable BioPython environment, A plain-text decision log, Test data or realistic sample material

Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using BioPython.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

RESPONSIBLE PRACTICE

This material is educational and does not provide medical advice or replace qualified clinical judgement.

PROJECT EVIDENCE

Document decisions another person can review

1. PURPOSE AND USER

State the intended user, need, expected outcome and constraints.

2. INPUTS AND ASSUMPTIONS

List source files, data, versions, permissions and assumptions.

PROJECT EVIDENCE

Tests, limitations and revision

3. NORMAL CASE

Expected result, observed result and evidence location.

4. BOUNDARY OR FAILURE CASE

What was difficult, what happened and why it matters.

5. REVISION AND LIMITATION

What changed, what improved and what remains unproven.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

FINAL DECISION

Is the project ready to share? State the evidence, remaining risk, owner and next review date.

Authoritative references

Global strategy on digital health 2020-2027

World Health Organization - reviewed 2026-08-21

<https://www.who.int/publications/i/item/9789240116870>

FHIR specification

HL7 International - reviewed 2026-08-21

<https://hl7.org/fhir/>

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the BioPython scope.